

Regime Consistency of an Entropy-Gradient Coupling: Ly α , BAO and RSD Tests within a Structural Coherence Framework

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WHAT THIS IS

- An empirical consistency test: does the same coupling function $T(z)$ match observations across three cosmological probes (Ly α P1D, BAO distances, RSD growth rates) without dataset-specific parameter adjustments?
- A structural coherence evaluation using the SCE framework: Layer Consistency, Model Plasticity, and Forbidden Pattern Coverage scored against Λ CDM.
- All tests use Λ CDM-calibrated background. Results demonstrate regime-consistency of the EFC coupling, not a fully self-consistent cosmology.

WHAT THIS IS NOT

- Not a full MCMC joint fit (channels tested independently; covariances between BAO and RSD not included).
- Not a theoretical derivation of the coupling (the functional form $T(z)$ is postulated, not derived from first principles).
- Not proof that EFC is correct (Λ CDM also fits the data; the question is structural discipline, not statistical knockout).

Abstract

This paper analyzes structural regime consistency rather than performing new parameter fits. We ask whether a single entropy-gravity coupling function $T(z)$, calibrated at $z \approx 0.5$ from BOSS DR12 distance ratios ($\alpha_{L2} = 0.34 \pm 0.16$), produces consistent predictions across three independent cosmological regimes without dataset-specific adjustments. At $z = 3$ (DESI Ly α P1D), EFC predicts a near-null effect ($T = 0.107$); data are consistent with null detection, with both amplitude and slope pulls below 0.5σ . At $z \approx 0.4\text{--}0.6$ (BOSS BAO + RSD), a one-parameter background modification ($\beta = 0.08$) simultaneously improves distance and growth fits over Λ CDM by $\Delta\chi^2 = -3.89$ on 9 data points. At $z \approx 1100$, $T(z) \approx 10^{-10}$ guarantees GR recovery. We evaluate both frameworks using the Structural Coherence Evaluation (SCE): EFC shows higher Layer Consistency (one mechanism across three regimes, no patches) and lower Model Plasticity (parameter count decreased from 3 to 1 across publications). Λ CDM retains higher Explanatory Compression (2.0 vs 1.3 in core scope). Explicit Forbidden Pattern declarations are tabulated in the Appendix. Neither framework dominates the full SCE profile.

1. Introduction

Energy-Flow Cosmology (EFC) proposes an entropy-gravity coupling that modifies gravitational dynamics at cosmological scales. The L2 (phenomenological cosmology) module parametrizes this coupling through a

transition function $T(z)$ derived from the matter fraction $\Omega_m(z)$, which naturally suppresses the effect at high redshift.

The central empirical question is not whether EFC provides a statistically superior fit to any single dataset — at current precision, Λ CDM fits most data within 1σ . Rather, the question is one of structural discipline: does a single coupling function, calibrated on one dataset, produce consistent predictions across multiple independent regimes without dataset-specific adjustments?

This paper tests that question against three probes spanning $z = 0.4$ to $z = 3$, and evaluates the result using the Structural Coherence Evaluation (SCE) framework — a set of five metrics designed to compare theoretical frameworks on structural grounds rather than pure goodness-of-fit.

2. The Coupling Function $T(z)$

The regime-transition function is defined as:

$$g_D(z) = (1 - \Omega_m(z)) / (1+z) \times \ln(1+z)$$

$$T(z) = g_D(z) / g_D(z_{\text{peak}}), \quad z_{\text{peak}} = 0.441$$

where $\Omega_m(z) = \Omega_m(1+z)^3/E^2(z)$ uses the Planck 2018 best-fit Λ CDM background ($h = 0.6736$, $\Omega_m = 0.3153$, $A_s = 2.1 \times 10^{-9}$, $n_s = 0.9649$). Key values:

z	$\Omega_m(z)$	$1-\Omega_m$	$g_D(z)$	$T(z)$	Regime
0.44	0.548	0.452	0.1066	1.000	Peak
0.51	0.613	0.387	0.1056	0.990	BOSS
2.33	0.945	0.055	0.0201	0.188	BAO Ly α
3.00	0.967	0.033	0.0114	0.107	Ly α P1D
1100	≈ 1.000	$\approx 10^{-6}$	$\approx 10^{-10}$	$\approx 10^{-10}$	CMB

Background calibration note: $T(z)$ is defined using Λ CDM-calibrated $\Omega_m(z)$. These tests therefore probe internal consistency of the EFC coupling across redshift regimes within a standard FRW background, not a fully self-consistent EFC cosmology.

3. Test 1: Ly α P1D at $z = 3$

3.1 What DESI Measures

DESI's Ly α flux P1D yields inferred linear matter power amplitude $\Delta_{\star}^2 = 0.379 \pm 0.032$ and slope $n_{\star} = -2.309 \pm 0.019$ at pivot $k = 0.009 \text{ s/km}$ ($k_{\text{com}} = 0.691 \text{ Mpc}^{-1}$ at $z = 3$), obtained via simulation-calibrated emulator marginalizing over IGM nuisance parameters (arXiv:2601.21432).

3.2 Λ CDM Baseline (CLASS 3.3.4)

Quantity	CLASS value	DESI observed
Δ_{LCDM}^2	0.3578	0.379 ± 0.032
n_{LCDM}	-2.3047	-2.309 ± 0.019
Amplitude pull	-0.66σ	$< 1\sigma$
Slope pull	+0.23σ	$< 1\sigma$

Λ CDM fits at $< 1\sigma$ on both quantities. Room for EFC is limited but nonzero: the amplitude is underpredicted by 0.66σ .

3.3 EFC Type A Prediction

Using $T(z=3) = 0.107$ and $\alpha_{L2} = 0.34$ (BOSS best-fit), the effective coupling at $z = 3$ is $A \cdot T = 0.037$. For $k_{tr} = 2.0 \text{ Mpc}^{-1}$, $p = 3$:

α_{L2}	$A \cdot T(z=3)$	Amplitude pull	Slope pull
0.18 (-1σ)	0.019	-0.45σ	$+0.11\sigma$
0.34 (best)	0.037	-0.27σ	$+0.01\sigma$
0.51 ($+1\sigma$)	0.055	-0.07σ	-0.09σ

Verdict: PASS. Both pulls $< 0.5\sigma$ across the full α_{L2} uncertainty range. EFC predicts $P1D \approx \Lambda\text{CDM}$ at $z = 3$; data are consistent with this null detection. This is a regime-consistency test: the effect is predicted to be present at $z \approx 0.5$ (measured) and absent at $z = 3$ (consistent with null).

4. Test 2: BAO Distances (BOSS DR12)

The EFC background modification enhances $H(z)$ at late times, reducing comoving distances:

$$H^2(a) = H_0^2 [\Omega_m \cdot a^{-3} + \Omega_\Lambda (1 + \beta \cdot T(a))]$$

We compute $D_M(z)/r_d$ and $D_H(z)/r_d$ via numerical integration ($N = 10,000$ steps, $r_d = 147.09 \text{ Mpc}$). Data: BOSS DR12 consensus (Alam et al. 2017).

z	D_M pull (EFC)	D_H pull (EFC)	D_M pull (ΛCDM)	D_H pull (ΛCDM)
0.38	$+0.29\sigma$	-1.10σ	$+1.15\sigma$	-0.53σ
0.51	-0.26σ	$+0.12\sigma$	$+0.67\sigma$	$+0.71\sigma$
0.61	$+0.14\sigma$	$+0.74\sigma$	$+1.17\sigma$	$+1.29\sigma$

$\chi^2_{\text{BAO}} = 1.93$ (EFC) vs **5.58** (ΛCDM). Improvement: $\Delta\chi^2 = -3.65$. ΛCDM systematically overpredicts D_M at all three redshifts; EFC corrects this via enhanced $H(z)$. All pulls $< 1.1\sigma$.

5. Test 3: RSD Growth Rate (BOSS DR12)

The same background modification that changes distances also modifies growth through increased Hubble friction. We solve the linear growth equation using RK4 (50,000 steps, $a = 10^{-4}$ to 1) with the modified $H(a)$. Note: no Poisson modification ($\mu = 1$); growth suppression comes entirely from background dynamics.

z	$f\sigma_8$ EFC	$f\sigma_8$ obs	σ	EFC pull	ΛCDM pull
0.38	0.464	0.497	0.045	-0.73σ	-0.46σ
0.51	0.463	0.458	0.038	$+0.13\sigma$	$+0.43\sigma$
0.61	0.460	0.436	0.034	$+0.71\sigma$	$+0.97\sigma$

Derived: $\sigma_{8,0} = 0.805$ (vs ΛCDM 0.811), $S_8 = 0.825$ (vs 0.832), growth suppression $\varepsilon_0 = 0.8\%$.

Verdict: PASS. $\chi^2_{\text{growth}} = 1.09$ (EFC) vs 1.34 (ΛCDM). EFC reduces the near- 1σ ΛCDM overprediction at $z = 0.61$. Growth suppression is in the correct direction for S_8 tension but addresses only $\sim 10\%$ of it.

6. Combined Result

Test	EFC χ^2	ΛCDM χ^2	$\Delta\chi^2$	Result
Ly α P1D ($z=3$)	$< 0.5\sigma$	$< 1\sigma$	—	Regime consistency
BAO (BOSS DR12)	1.93	5.58	-3.65	EFC improved
RSD (BOSS DR12)	1.09	1.34	-0.25	Marginal improvement
BAO + RSD combined	3.02	6.92	-3.89	$\sim 2\sigma$ preference

The combined improvement is $\Delta\chi^2 = -3.89$ for one additional parameter (β). For 1 DOF, $\Delta\chi^2 = 3.84$ corresponds to the 95% CL boundary. This is suggestive but not conclusive. The key structural point is that

both geometry and growth improve from the same single modification — no separate tuning per dataset.

6.1 Regime Coherence Summary

Regime	z	T(z)	EFC prediction	Data	Result
Growth	0.4–0.6	0.95–1.0	Enhanced H, suppressed growth	BAO + f σ_8	PASS
Ly α	3.0	0.107	Near-null effect	$\Delta^2_{\star}, n_{\star}$	PASS
CMB	1100	$\approx 10^{-10}$	GR recovery	Envelope	PASS

Note: The $z = 3$ and $z = 1100$ passes follow from the $T(z) \propto (1 - \Omega_m(z))$ structure. They are consequences of the ansatz, not independent confirmations. The structurally non-trivial result is the dual improvement at $z \approx 0.5$ (BAO + RSD from same β).

7. Structural Coherence Evaluation (SCE)

The SCE framework evaluates theoretical frameworks on five structural metrics. We apply it to both EFC (L2) and Λ CDM using locked decisions D1–D5 from the Compression Decision Record v1.0.

7.1 Explanatory Compression

Framework	Primitives	FDOF	Phenomena	C (core)	C (w/ tensions)
Λ CDM	2 (Λ , CDM)	2	4	2.0	1.7
EFC (L2)	3 (S, μ , T)	1	4	1.3	1.7–2.0

Λ CDM has higher compression in core scope. Roughly equal when tensions are included. Ranking reverses under alternative decision D2a (T(z) fully derived). *Not decisive*.

7.2 Anomaly Burden

Framework	A1	A2	A3	Burden type
Λ CDM	3	2	~ 1	Precision tensions ($S_8, H_0, w_0 w_a$)
EFC (L2)	0	6	3	Theoretical incompleteness

EFC's dominant A1 anomalies (CMB, BBN) were reclassified to A2/A3 after the high- z envelope demonstrated margins of $10^8\times$ (CMB) and $10^{23}\times$ (BBN) below observational bounds. **Draw with asymmetric failure modes.**

7.3 Model Plasticity

Indicator	Λ CDM	EFC (L2)	Winner	Note
Variant proliferation	High	Moderate	EFC	
Parameter growth	Growing	Decreasing	EFC	$3 \rightarrow 1$ FDOF
Special-case rules	Some	None	EFC	
Ad hoc patches	Several	None	EFC	P1D: no new params

EFC has lower plasticity. Key evidence: FDOF count decreased from $3 \rightarrow 1$ across publications; the P1D and BAO+RSD tests added no new structure. Caveat: EFC is younger, so low plasticity partly reflects limited testing.

7.4 Layer Consistency

Evidence	What it shows	Patches needed
BOSS $z=0.5$: $\alpha_{L2} = 0.34 \pm 0.16$	Coupling measured at peak	None
BOSS BAO+RSD: $\beta = 0.08$	Background channel tested	None
DESI P1D $z=3$: pulls $< 0.5\sigma$	Consistent with predicted null	None
CMB $z \approx 1100$: $T \approx 10^{-10}$	GR recovery	None (by construction)

Λ CDM's cross-regime tensions (S_8, H_0) suggest mechanism inconsistency. **EFC advantage** — with the calibration caveat that all tests use Λ CDM background.

7.5 Forbidden Pattern Coverage

EFC has five explicitly declared, testable forbidden patterns (see Appendix A). None are triggered by current data. Λ CDM has not declared comparable forbidden patterns in the SCE sense. **EFC advantage (early).**

8. Complete SCE Profile

SCE Metric	Λ CDM	EFC (L2)	Winner
Compression	2.0	1.3	Λ CDM (core)
Anomaly Burden	3 $\times$ A1, 2 $\times$ A2	0 $\times$ A1, 6 $\times$ A2	Draw (asymmetric)
Plasticity	High / growing	Low / stable	EFC
Layer Consistency	Cross-regime tensions	3 regimes consistent	EFC
Forbidden Patterns	Not declared	5 tested, 0 triggered	EFC (Appendix)

Neither framework dominates the full SCE profile. Λ CDM wins on raw compression; EFC wins on structural discipline. Compression alone cannot adjudicate between these frameworks at their current maturity levels.

9. Open Requirements

- Full MCMC joint fit:** Proper covariance matrices between BAO and RSD within BOSS. Without this, $\Delta\chi^2 = -3.89$ is indicative, not definitive.
- Neutrino mass degeneracies:** Would shift the σ_8 landscape and potentially affect growth constraints.
- Non-linear regime:** All analysis is in the linear regime. Scale-dependent effects (L3) are excluded per decision D4a.
- Self-consistent EFC background:** $T(z)$ uses Λ CDM $\Omega_m(z)$. A fully self-consistent analysis requires deriving $H(z)$ from EFC field equations.
- CMB power spectrum:** The high- z envelope guarantees compatibility but does not replace a full Boltzmann calculation.
- First-principles β :** $\beta = 0.08$ is fitted, not derived. Its relationship to $\alpha_{L2} = 0.34$ is an open theoretical question (addressed in companion paper).

10. Conclusion

A single coupling function $T(z)$, calibrated at $z \approx 0.5$, produces consistent predictions across three cosmological regimes without dataset-specific adjustments. A one-parameter background modification ($\beta = 0.08$) simultaneously improves BAO distances and growth rates over Λ CDM by $\Delta\chi^2 = -3.89$ on 9 data points. The high- z null prediction at $z = 3$ is consistent with DESI Ly α P1D data. GR recovery at recombination is guaranteed by $T(z) \rightarrow 0$ with margins exceeding 10^8 .

Evaluated within the SCE framework, EFC shows higher Layer Consistency (one mechanism, three regimes, no patches) and lower Model Plasticity (decreasing parameter count). Λ CDM retains higher Explanatory Compression in core scope. Forbidden Pattern declarations are tabulated in the Appendix. Neither framework dominates the full structural profile.

The critical path forward is not more structural evaluation but concrete theoretical completion: a self-consistent EFC background, full Boltzmann-level CMB predictions, and a first-principles derivation of β . These are addressed in a companion paper on the theoretical regime architecture.

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Appendix A: Declared Forbidden Patterns

The following patterns are declared in advance as conditions under which EFC's regime-transition logic would be internally inconsistent. They serve as explicit falsifiability criteria within the SCE framework.

Pattern ID	Definition	Status
EFC-FP-P1D	If $T(z=3)/T(\text{peak}) > 0.5$ AND ($ \text{amplitude pull} > 3\sigma$ OR $ \text{slope pull} > 3\sigma$), then regime-transition logic is inconsistent.	Not triggered
EFC-FP-HIGHZ	If Boltzmann solver with $\mu(z) = 1 + A \cdot T(z)$ produces CMB C_l residuals exceeding Planck error bars.	Not triggered
EFC-FP-RSD	If $f\sigma_8$ deviates $> 3\sigma$ at any BOSS z while α_{L2} within $\pm 2\sigma$ of best-fit.	Not triggered
EFC-FP-BAO	If $\beta > 0$ simultaneously worsens both BAO and $f\sigma_8$ vs ΛCDM .	Not triggered
EFC-FP-4CH	If β passes BAO+RSD but fails CMB lensing or SN Ia $> 3\sigma$.	Not triggered

Note: These patterns are pre-registered falsifiability conditions, not post hoc interpretations. Pattern EFC-FP-P1D was declared in the kill-test specification (v1.3) before the Type A prediction was computed. Patterns EFC-FP-BAO and EFC-FP-4CH were declared in the present work.